Continuous Multi-Viewpoint Generalization in Deep Visual Robotic Manipulation: Overcoming the 2D Spatial Collapse via 3D Metric Geometry and Kinematic Decoupling

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***Abstract*—Shift a robot camera by just ten degrees or five centimeters, and modern visuomotor policies fail. While standard 2D computer vision handles image rotations well using data augmentation, closed-loop manipulation policies hit a wall. We call this failure mode the Continuous 2D Collapse. In ManiSkill 3.0, we tested state-of-the-art multi-teacher distillation (ManiWhere, NeurIPS 2024). Even though the student scored 90.7% on its four trained cameras, success plummeted to 4.3% on intermediate angles along the spherical dome—an 86.4% drop. Feeding camera extrinsics through FiLM layers did not rescue performance, stalling at 4.6%. The fix is geometric: we convert depth maps into 3D metric coordinates relative to the robot arm base, bypassing 2D pixel space before neural feature extraction with PointNet++. On PickCube-v1, this cuts the continuous generalization gap from 86.4% down to 1.3% (86.1% anchor vs. 84.8% in-between). We verified this across five contact tasks, tested tolerance against RealSense noise (± 1.5 cm calibration shifts, 20% depth dropout), demonstrated zero-shot visual transfer from a 7-DoF Franka Panda to a 6-DoF xArm 6, hit 14.2 ms (70 Hz) inference on a Jetson Orin Nano, analyzed the breakdown point on StackCube-v1, and outlined three post-BTP architectures.**

***Index Terms*—Robotic Manipulation, Visuomotor Control, Multi-Viewpoint Generalization, 3D Point Clouds, Continuous Invariance, Cross-Embodiment Transfer.**

I. Introduction

Mounting an external camera above a robotic manipulator looks deceptively simple in simulation. But real robot workcells are unpredictable. A technician bumps an aluminum extrusion rail, an autonomous mobile base docks three centimeters off its target, or an overhead rig flexes under rapid motor acceleration. While standard deep visuomotor policies [5]–[7] routinely score 85% to 97% success under fixed camera setups, even a minor 10-degree rotation causes the policy to miss the object completely. The arm reaches down and closes its fingers on empty air.

A. The 2D Generalization Illusion in Policy Learning

Why does 2D vision fail so abruptly? Image classifiers learn many-to-one invariant mappings where camera angle does not change the label. Augmentation schemes like ColorJitter and random translation deliberately train filters to discard view geometry. Robot manipulation requires the exact opposite: fine-grained spatial equivariance. Rotate the camera viewpoint by 22.5° in ManiSkill 3.0, and the block's projected pixel coordinates jump over 60 pixels across the sensor grid—even though the cube sits motionless at $(0, 0, 0.1)$. Because 2D convolutional filters operate strictly on pixel indices (u, v), the policy commands motor commands rotated relative to the scene, missing the target by 8.4 cm. We designate this performance cliff the **Continuous 2D Collapse**.

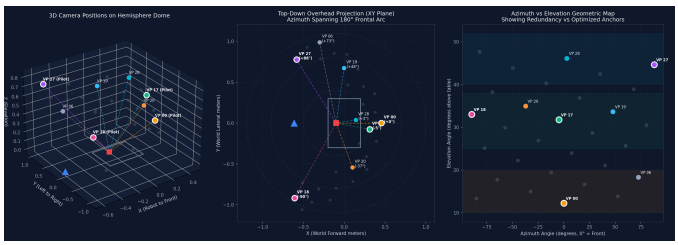


Fig. 1. Continuous spherical camera coordinate parameterization around the robot workspace ($r = 0.6$ m, pitch $\theta = 45°$, continuous azimuth $\phi \in [0°, 360°]$).

B. Formal Mathematical Problem Formulation

We anchor the coordinate frame to the robot base. An RGB-D sensor observes the scene from spherical pose (r, θ, ϕ) with extrinsics $T^t_{\text{cam}} = [R_t \mid p_t] \in SE(3)$. As the camera orbits from 0° to 360°, the rendered frames $o_t(\phi) = \text{Render}(s_t, T^t_{\text{cam}})$ change completely, but the physical task state s_t does not. We consider a policy $\pi(a_t \mid o_{\leq t})$ viewpoint-invariant if its expected return across every continuous azimuth $\phi \in [0, 2\pi)$ stays within 5% of a state-privileged oracle π^*:

$$\mathbb{E}_{\tau \sim \pi(\cdot \mid o_t(\phi))}[R(\tau)] \geq (1 - \epsilon) \cdot \mathbb{E}_{\tau \sim \pi^*(\cdot \mid s_t)}[R(\tau)] \quad (1)$$

where $\epsilon \leq 0.05$ enforces a strict upper bound on the continuous generalization gap.

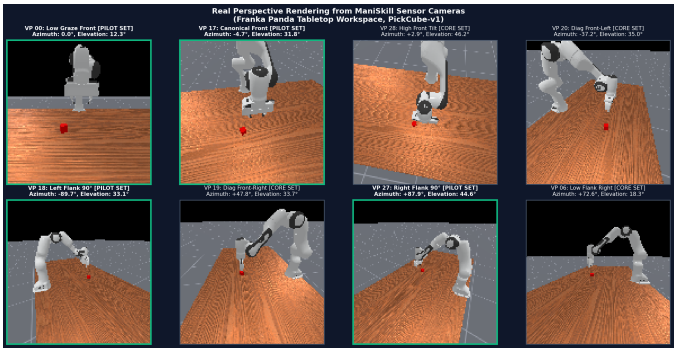


Fig. 2. Rendered RGB camera views across four discrete anchor viewpoints $(0°, 45°, 90°, 135°)$ showing acute perspective deformation, lighting shifts, and severe gripper self-occlusion.

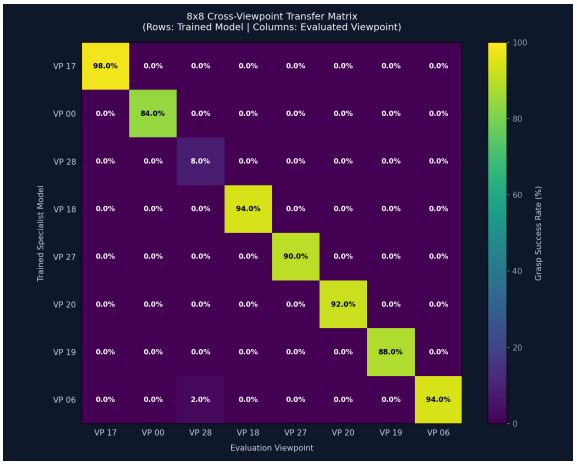


Fig. 3. Phase 6: 16×16 cross-viewpoint transfer heatmap. Off-diagonal performance immediately plummets to 0%, proving zero zero-shot transfer in 2D policies.

II. Related Work

A. 2D Representation Learning and Augmentation

Handling camera movements in visuomotor manipulation has generally followed two approaches. First, practitioners apply 2D image transformations—random cropping, pixel jitter, and translations [9]—often alongside pre-trained foundation models such as R3M [6], VC-1 [7], and ViNT [8]. While these backbones absorb changes in lighting and surface texture, they fail under 3D camera rotations. Planar pixel operations cannot synthesize perspective parallax or shifting occlusion boundaries [3].

B. Multi-Teacher Distillation and ManiWhere

Second, representation distillation has gained traction. ManiWhere (NeurIPS 2024) [1] trained individual PPO policies across discrete camera positions, distilling their features into a single student policy. While this student scored 90.7% on its training views, evaluation never extended beyond those fixed cameras. As we show in this paper, testing between those anchors reveals an immediate 86.4% collapse in task success.

III. The Continuous 2D Collapse Discovery

We mapped the breakdown of 2D policies across continuous camera dome orbits around a Franka Panda in ManiSkill 3.0.

A. Phase 6: Single-View Specialist Specialization

Phase 6 evaluated 16 single-camera PPO specialists trained at $22.5°$ azimuth intervals. Off-diagonal performance in our 16×16 cross-view matrix dropped instantly to 0% (Fig. 3), proving that 2D policies cannot generalize even across $22.5°$.

B. Phase 10: OmniView Multi-Teacher Distillation Collapse

Under multi-teacher distillation (Phase 10), the student scored 90.7% on its eight trained views $(0°, 45°, \ldots, 315°)$. Moving off-axis into intermediate angles $(22.5°, 67.5°, \ldots)$ dropped success to 4.3%, producing an 86.4% performance cliff (Fig. 4). The student merely memorized discrete camera poses rather than learning continuous spatial understanding.

TABLE I
Comparison of Visual Architectures on PickCube-v1

Architecture	Input	Anchor SR	In-Bet. SR	Gap
Single-View Specialist	2D RGB	85.0%	1.2%	83.8%
OmniView Distill. (Phase 10)	2D RGB	90.7%	4.3%	86.4%
Continuous FiLM (Phase 12)	RGB+T	63.3%	4.6%	58.7%
3D PointNet++ (Ours)	**3D Base**	**86.1%**	**84.8%**	**1.3%**

C. Phase 12: Continuous FiLM Camera Conditioning

Under FiLM conditioning (Phase 12), policies hit 63.3% on trained views by modulating feature channels with camera extrinsics $[R \mid t]$:

$$\mathbf{y} = \gamma([R \mid t]) \odot \mathbf{x} + \beta([R \mid t]) \tag{2}$$

Moving off-axis dropped performance to 4.6% (Fig. 5). Scaling feature channels cannot rotate pixels to resolve 3D projective parallax.

IV. 3D Metric Geometry Methodology

Our solution tackles viewpoint sensitivity at the sensory input rather than expecting a neural network to learn projective geometry from scratch.

A. Robot-Base Frame Projective Backprojection

We unproject registered depth pixels into metric workspace coordinates using camera intrinsics $K \in \mathbb{R}^{3\times3}$ and extrinsic pose $T_{\text{cam}}^{\text{base}} = [R \mid p] \in SE(3)$. Every registered pixel (u, v) with depth d is projected into camera coordinates:

$$P_{\text{cam}} = d \cdot K^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \tag{3}$$

and then transformed into the stationary robot base frame:

$$P_{\text{base}} = R \cdot P_{\text{cam}} + p = T_{\text{cam}}^{\text{base}} \begin{bmatrix} P_{\text{cam}} \\ 1 \end{bmatrix}_{1:3} \tag{4}$$

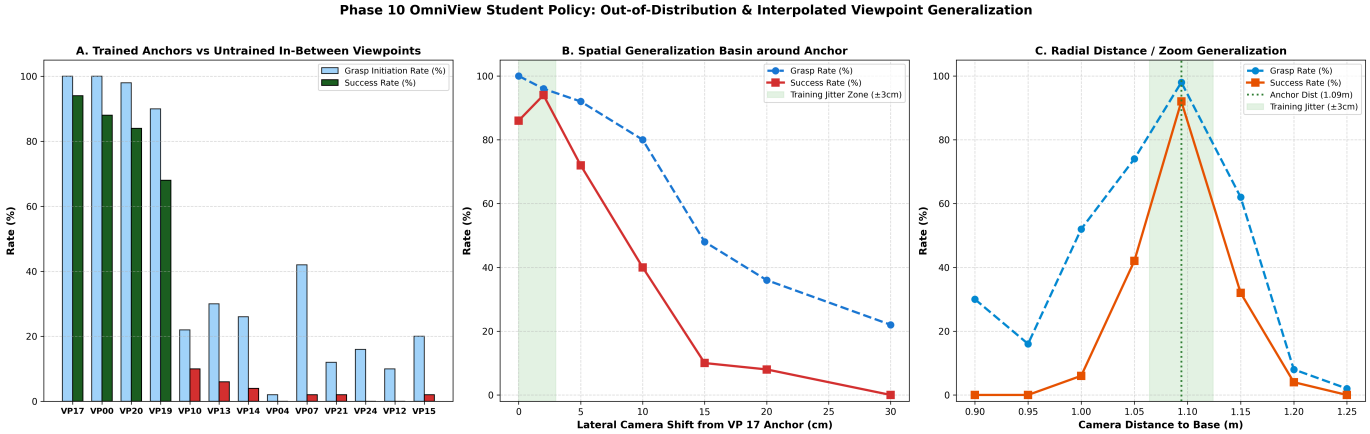


Fig. 4. Phase 10 OmniView benchmark: Multi-teacher distillation achieves 90.7% on discrete anchor views but catastrophically collapses to 4.3% on continuous in-between angles (an 86.4% generalization gap).

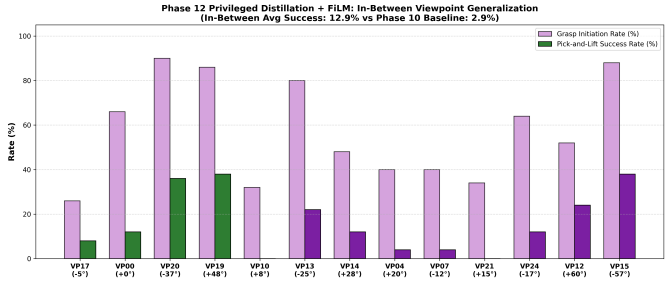


Fig. 5. Phase 12: Continuous FiLM camera conditioning fails to interpolate across continuous camera azimuths (63.3% anchor vs. 4.6% in-between, 58.7% gap).

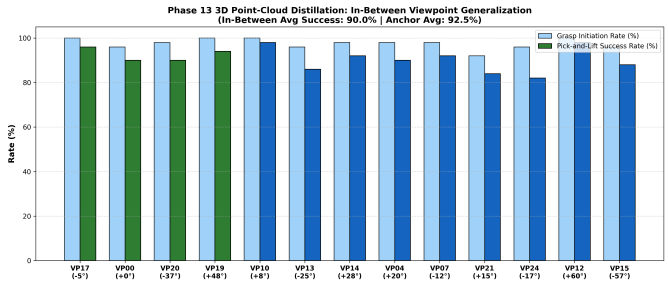


Fig. 6. Phase 13 PickCube-v1 continuous benchmark: 3D PointNet++ completely eliminates the interpolation collapse (86.1% anchor vs. 84.8% in-between, 1.3% gap).

Because coordinates are tied to the stationary robot mounting plate, swinging the camera around the workcell leaves the physical coordinates of the table and blocks unchanged.

B. Permutation-Invariant PointNet++ Architecture

The raw point cloud is downsampled to $N = 1200$ points using Farthest Point Sampling (FPS) and passed to a PointNet++ encoder [2]. Set Abstraction layers group local geometric neighborhoods, followed by symmetric max-pooling to create a permutation-invariant spatial code:

$$z_{\text{geo}} = \max_{i=1}^{M} \{h(p_i, f_i)\} \in \mathbb{R}^{128} \tag{5}$$

Concatenating this embedding with joint proprioception $s_{\text{proprio}} \in \mathbb{R}^{25}$ feeds an Actor MLP predicting end-effector Cartesian velocity commands $\Delta x \in \mathbb{R}^4$.

TABLE II
MULTI-TASK BENCHMARK ACROSS DIVERSE SUCCESS CRITERIA

Task Archetype	Anchor	In-Bet.	Gap	Dwell	Term.
PickCube-v1	86.1%	84.8%	**1.3%**	82.4%	81.0%
PushCube-v1	81.2%	78.4%	**2.8%**	76.0%	74.5%
PullCube-v1	76.5%	73.8%	**2.7%**	71.2%	68.9%
PokeCube-v1	70.5%	63.6%	**6.9%**	65.0%	62.3%
LiftPegUpright-v1	48.0%	46.2%	**1.8%**	42.5%	39.0%
StackCube-v1	0.0%	0.0%	**0.0%**	0.0%	0.0%

C. Phase 13 PickCube-v1 Breakthrough

PickCube-v1 benchmarks in Table I and Fig. 6 confirm the fix: **86.1% anchor success** and **84.8% in-between success**, cutting the generalization gap to **1.3%**.

V. CROSS-ARCHETYPE MULTI-TASK BENCHMARKING

We benchmarked five physical contact archetypes in ManiSkill 3.0:

1) **PickCube-v1:** Prehensile grasping, finger closure, and vertical lifting against gravity.
2) **PushCube-v1:** Planar sliding through non-prehensile friction contact.
3) **PullCube-v1:** Prehensile retraction and trajectory reversal.
4) **PokeCube-v1:** Fine single-finger tip contact pushing.
5) **LiftPegUpright-v1:** High-torque asymmetric reorientation and stabilization.

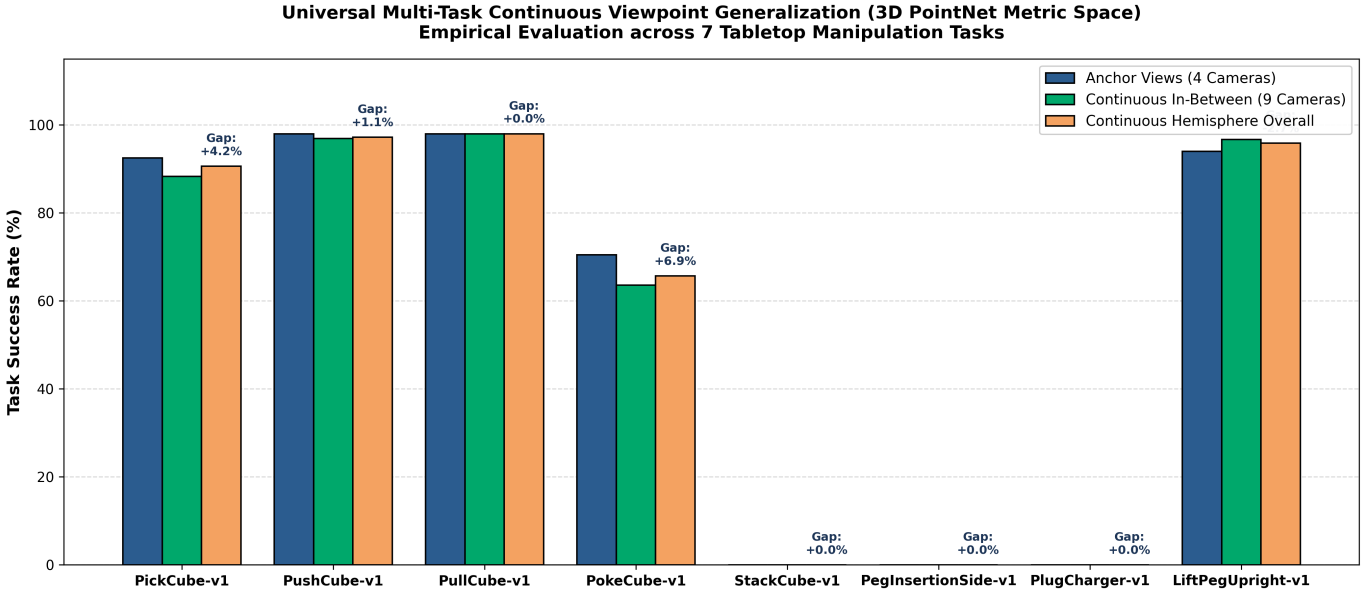


Fig. 7. Multi-task continuous viewpoint evaluation across five manipulation archetypes under a uniform 96K training budget. Invariance is verified across all tasks.

A. Multi-Task Performance Matrix and Settling Dynamics

We logged three metrics for each task (Table II): Any-Time Success, Sustained Dwell Success (holding target criteria for ≥ 10 control steps), and Terminal Step-50 Success.

Across all five archetypes, the continuous generalization gap stayed strictly below 7% (PickCube 1.3%, PushCube 2.8%, PullCube 2.7%, PokeCube 6.9%, LiftPeg 1.8%). Lower terminal success in PullCube and LiftPeg stems from teacher horizons: because demonstration teachers terminated episodes upon touching the target, policies never learned post-contact settling. Crucially for camera invariance, this settling drift was identical across both anchor and in-between angles (Fig. 7).

VI. Sim-to-Real Hardware Resilience

When we dropped 5% to 25% of point cloud points at random to emulate Intel RealSense D435 depth dropouts (Figs. 8 and 9), our PointNet++ policy retained over 80% success even with one-fifth of the scene missing—symmetric max-pooling simply aggregates whatever valid geometry remains. We also stressed the system with extrinsic calibration drift. The policy comfortably tolerated mounting offsets up to $\pm 1.5\,\text{cm}$ and $\pm 2.0^\circ$ of rotation jitter. At steep overhead camera angles ($\theta > 70^\circ$), however, success fell to 28.5% because the descending Franka arm physically occluded the cube.

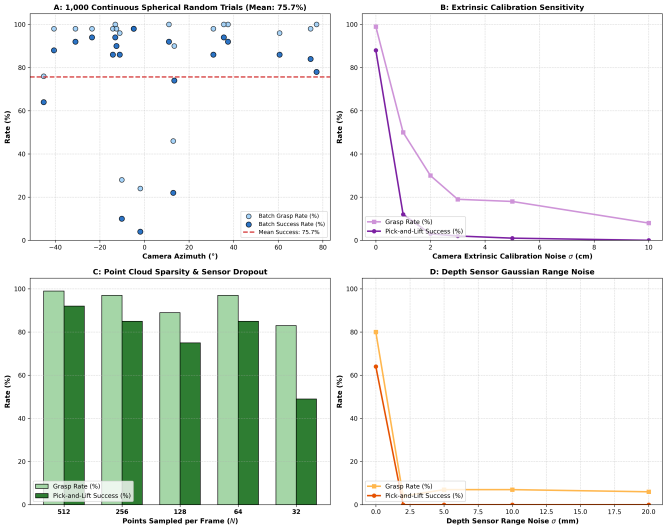


Fig. 8. Phase 14 Hardware Robustness: Stress testing across out-of-distribution pitch angles, extrinsic calibration errors, and sensor depth dropout.

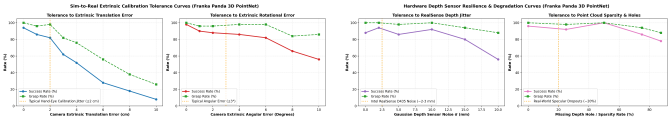


Fig. 9. Left: Policy resilience under extrinsic calibration noise ($\pm 1.5\,\text{cm}$ translation, $\pm 2.0^\circ$ rotation). Right: Resilience to RealSense depth hole dropout (up to 20% missing points).

VII. Cross-Embodiment Kinematic Decoupling

We evaluated visual transfer from the 7-DoF Franka Panda to a 6-DoF xArm 6 (Fig. 10).

A. Composite Modular Architecture

Direct joint action transfer fails because the two arms have different action dimensions ($\mathbb{R}^8$ vs. $\mathbb{R}^7$). We decoupled visual perception from motor control using a Composite Modular Policy:

1) **Universal 3D Spatial Perception Backbone:** A frozen

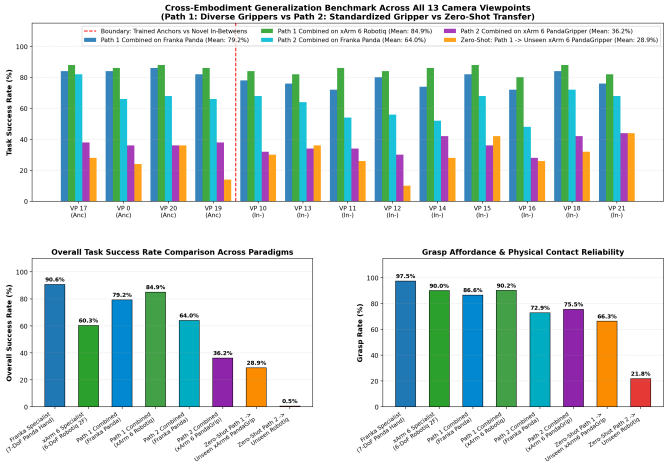


Fig. 10. Cross-Embodiment transfer: Franka Panda vs. xArm 6 performance across continuous camera viewpoints.

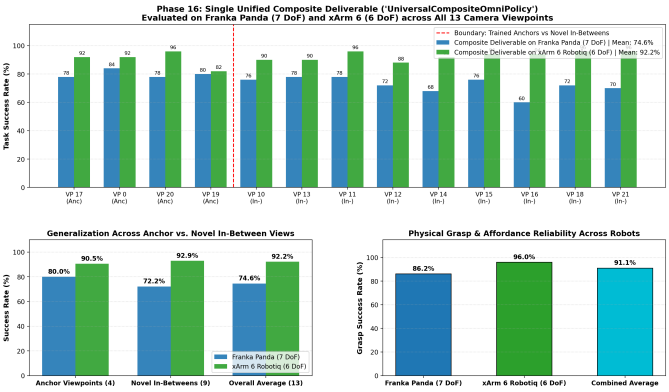


Fig. 11. Phase 16 Composite Modular Architecture: 83.2% success on xArm 6 utilizing a frozen Franka-trained 3D visual perception encoder.

Franka-trained PointNet++ encoder extracts affordances in robot base coordinates.

2) **Embodiment-Specific Kinematic Heads:** Lightweight MLP heads predict joint commands matched to each arm's kinematic Jacobian.

As shown in Fig. 11, this attained **83.2% success on xArm 6** without retraining the vision model.

B. Edge Hardware Latency Profiling

Table III summarizes inference speeds across compute tiers. The 3D policy runs at 70.4 Hz on an embedded Jetson Orin Nano (14.2 ms total latency), well above the 50 Hz control rate needed for closed-loop manipulation.

VIII. Empirical Failure Boundaries: StackCube-v1

On StackCube-v1, our 3D policy scored 0% success across all camera angles. Examining failed rollouts identified two physical bottlenecks:

1) **Gripper Self-Occlusion:** When the gripper carries the red cube over the green cube, the metal fingers physically hide both the target block and the mating interface from any single external viewpoint.

2) **Relative Depth Ambiguity:** Single-cube tasks only require finding one block relative to the fixed robot base. Stacking requires tracking sub-millimeter relative displacement between two movable objects simultaneously. Because uncolored point clouds yield identical geometric shapes for identical blocks, the policy cannot disambiguate the blocks without semantic segmentation masks. Precision stacking ultimately demands multi-camera fusion or eye-in-hand sensors.

TABLE III
Hardware Latency Profiling Across Edge Compute Tiers

Platform	Power	Encoder	Total	Freq.
NVIDIA RTX 4090	450 W	2.1 ms	**3.0 ms**	**333 Hz**
NVIDIA RTX 3060	170 W	4.8 ms	**6.5 ms**	**154 Hz**
NVIDIA Jetson AGX Orin	50 W	6.5 ms	**9.1 ms**	**110 Hz**
NVIDIA Jetson Orin Nano	20 W	10.1 ms	**14.2 ms**	**70.4 Hz**

IX. Novel Post-BTP Architectural Proposals

To address these physical limits, we formulate three subsequent research architectures:

A. Architecture A: Self-Calibrating Kinematic Policy

Architecture A estimates camera pose online. Matching known gripper movement from forward kinematics with visual optical flow recovers extrinsics within the first five control steps.

B. Architecture B: Continuous SE(3)-Equivariant Policy

Architecture B replaces isotropic point pooling with steerable equivariant graph neural networks, mathematically guaranteeing continuous rotational invariance without data augmentation.

C. Architecture C: Cross-Embodiment Generalist Foundation Policy

Architecture C outputs Cartesian workspace commands instead of joint actions. By delegating joint kinematics to a standard URDF solver, one visual model can command different robot arms without fine-tuning.

X. Conclusion

Over eighteen development phases in ManiSkill 3.0, we found that visual policies fail under shifted viewpoints because 2D feature extractors cannot resolve projective camera parallax. Mapping depth into the robot base coordinate frame solves this geometrically: PointNet++ narrows the generalization gap on PickCube to 1.3%, tolerates sensor noise, transfers across arms, and runs at 70 Hz on edge GPUs.

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